

# Homework 8

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1. Let  $X, Y$  be normed vector spaces. Show that  $T : X \rightarrow Y$  is bounded if and only if it maps bounded sets of  $X$  to bounded sets of  $Y$ .

*Proof.* If  $T$  is bounded, then let  $A$  be a bounded set of  $X$ , and let  $d$  be the diameter of  $A$ . For each point  $a, b \in A$ , we have  $|b - a| \leq d$ , and thus  $|T(b) - T(a)| \leq \|T\| \cdot |b - a| \leq \|T\| \cdot d$ , so  $T(A)$  is bounded with diameter at most  $\|T\| \cdot d$ .  $\square$

2. Let  $T$  be a bounded, surjective linear operator from a normed vector space  $X$  to a normed vector space  $Y$ . Suppose there is a positive  $b$  such that

$$|T(u)|_Y \geq b|u|_X$$

for all  $u \in X$ . Show that  $T^{-1} : Y \rightarrow X$  exists and is bounded.

*Proof.* To show that  $T^{-1}$  exists, we first need to show that  $T$  is injective. Indeed, if  $u, v \in X$  and  $u \neq v$ , then  $|u - v|_X > 0$ . Thus

$$|T(u) - T(v)|_Y = |T(u - v)|_Y \geq b|u - v|_X > 0,$$

so  $T(u) \neq T(v)$ .

We should also show that  $T^{-1}$  is a linear operator. Let  $v_1, v_2 \in Y$ . Then  $v_1 = T(u_1)$  and  $v_2 = T(u_2)$  for some  $u_1, u_2 \in X$ . Then given scalars  $c_1$  and  $c_2$ , we have

$$\begin{aligned} T^{-1}(c_1 v_1 + c_2 v_2) &= T^{-1}(c_1 T(u_1) + c_2 T(u_2)) \\ &= T^{-1}(T(c_1 u_1 + c_2 u_2)) \\ &= c_1 u_1 + c_2 u_2 \\ &= c_1 T^{-1}(v_1) + c_2 T^{-1}(v_2) \end{aligned}$$

as desired.

Finally, we show that  $T^{-1}$  is bounded. Let  $v \in Y$ . Then

$$|T^{-1}(v)|_X \leq \frac{1}{b} \cdot |T(T^{-1}(v))|_Y = \frac{1}{b} \cdot |v|_Y.$$

Since this is true for all  $v \in Y$ ,  $T^{-1}$  is bounded.  $\square$

3. Let  $C_c[1, \infty)$  be the set of all continuous functions defined on  $[1, \infty)$  that are compactly supported, endowed with the  $C^0$ -norm. Here recall that the function  $f$  is compactly supported if the closure of the set  $\{x \in [1, \infty) : f(x) \neq 0\}$  is compact. Define  $T : C_c[1, \infty) \rightarrow C_c[1, \infty)$  as

$$Tf(x) = \frac{f(x)}{x}.$$

Show that  $T$  is a bounded linear operator, and  $T^{-1}$  exists but is not bounded.

*Proof.* First, we have

$$\begin{aligned} \|Tf(x)\|_{C^0} &= \left\| \frac{f(x)}{x} \right\|_{C^0} \\ &= \sup_{x \in [1, \infty)} \left| \frac{f(x)}{x} \right| \\ &\leq \sup_{x \in [1, \infty)} |f(x)| \\ &= \|f(x)\|_{C^0} \end{aligned}$$

since  $|f(x)/x| \leq |f(x)|$  for all  $x \in [1, \infty)$ . Thus  $\|T\|$  is at most 1, so  $T$  is bounded.

Now define

$$T^{-1}f(x) = xf(x).$$

The function  $T^{-1}f$  is in fact compactly supported since if  $f(x) = 0$ ,  $xf(x) = 0$ , and since  $x \neq 0$ ,  $xf(x) \neq 0$  when  $f(x) \neq 0$ , so  $T^{-1}f$  has the exact same support as  $f$ . We have

$$\begin{aligned} T^{-1}(Tf(x)) &= x \cdot \frac{f(x)}{x} = f(x) \\ \text{and } T(T^{-1}f(x)) &= \frac{xf(x)}{x} = f(x), \end{aligned}$$

so  $T^{-1}$  is in fact the inverse of  $T$ . It is also linear because

$$\begin{aligned} T^{-1}(cf(x) + dg(x)) &= xcf(x) + xdg(x) \\ &= cT^{-1}(f(x)) + dT^{-1}(g(x)). \end{aligned}$$

Now consider the set of functions

$$f_n(x) = \begin{cases} nx - n^2 & n \leq x \leq n+1 \\ -nx + n^2 + 2n & n+1 \leq x \leq n+2 \\ 0 & \text{elsewhere} \end{cases}$$

where  $n \in \mathbb{Z}^+$ . These are spikes with peak at  $x = n+1$ , where  $f_n(n+1) = n$ . Then  $f_n$  is compactly supported (with support  $[n, n+2]$ ) and continuous, so  $f_n \in C_c[1, \infty)$  for all  $n$ . However,

$$\begin{aligned} \frac{\|T^{-1}f_n(x)\|_{C^0}}{\|f_n(x)\|_{C^0}} &= \frac{\sup_{x \in [1, \infty)} |xf_n(x)|}{\sup_{x \in [1, \infty)} |f_n(x)|} \\ &= \frac{(n+1) \cdot n}{n} \\ &= n+1. \end{aligned}$$

Since this is true for any  $n$ ,  $T^{-1}$  is unbounded. □

4. Define  $T : C[0, 1] \rightarrow C[0, 1]$  (both have  $C^0$  norm) as

$$Tf(x) = \int_0^x f(s) \, ds.$$

Find  $\mathcal{R}(T)$  (the range of  $T$ ) and  $T^{-1} : \mathcal{R}(T) \rightarrow C[0, 1]$ . Is  $T$  bounded? How about  $T^{-1}$ ?

*Proof.* The inverse  $T^{-1}$  is the operator

$$T^{-1}f = \frac{df}{dx},$$

which we have already mentioned in class is linear. This inverse only works as such, however, if  $f$  is differentiable, which suggests that  $\mathcal{R}(T)$  is likely to have something to do with  $C^1[0, 1]$ : the set of differentiable functions on  $[0, 1]$ . We need to restrict, however, to  $D = \{f \in C^1[0, 1] : f(0) = 0\}$  since the constants of integration will be taken out. Indeed, if  $f \in D$ , then  $f = \int_0^x \frac{df}{ds}(s) \, ds$  by the fundamental theorem of calculus, so  $f \in \mathcal{R}(T)$  and  $D \subset \mathcal{R}(T)$ . Also by the fundamental theorem of calculus, we get that  $\int_0^x f(s) \, ds$  is differentiable for all  $f \in C[0, 1]$ , so  $\mathcal{R}(T) \subset D$ . Thus  $\mathcal{R}(T) = D$ . Note that in this proof, we have used the fundamental theorem of calculus twice, showing that  $T^{-1}$  is in fact an inverse of  $T$ .

We now need to think about boundedness of  $T$  and  $T^{-1}$ . First, for any  $x \in [0, 1]$ ,

$$\begin{aligned} |Tf(x)| &= \left| \int_0^x f(s) \, ds \right| \\ &\leq \int_0^x |f(s)| \, ds \\ &\leq \int_0^x \|f(s)\|_{C^0} \, ds \\ &= x \|f\|_{C^0} \\ &\leq \|f\|_{C^0}. \end{aligned}$$

Thus we can take the supremum of this inequality over all  $x \in [0, 1]$  and get

$$\|Tf\|_{C^0} \leq \|f\|_{C^0},$$

so  $T$  is bounded.

On the other hand, consider  $f = \sin(nx) \in D$  for  $n \in \mathbb{Z}^+$ . Then  $\|f\| \leq 1$ , but

$$\|T^{-1}f\| = \|n \cos(nx)\| = n$$

since  $n \cos(nx) \leq n$  and  $n \cos(n \cdot 0) = n$ . Then

$$\frac{\|T^{-1}f\|}{\|f\|} \geq n$$

for each  $n$ , so  $T^{-1}f$  is unbounded. □